

Advanced Topics: AI Flywheels

Lecture 14
CS 155 (Winter 2026)

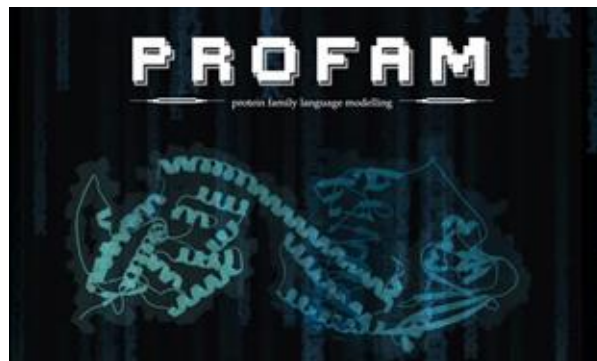
Topics for Today

Basic Modern Concepts

AI Flywheels

Case Studies

Foundation Models are Everywhere



Simple Ways to Use Foundation Models

Supervised Learning



Decoder

Embedding



Pretraining



Background
Dataset

Supervised Learning



LLaMA

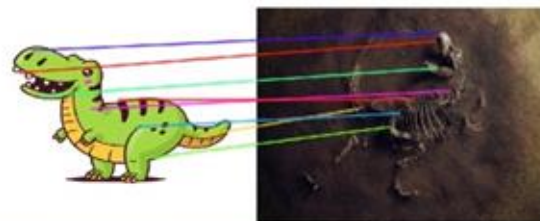
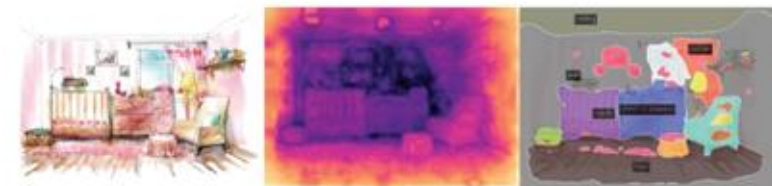
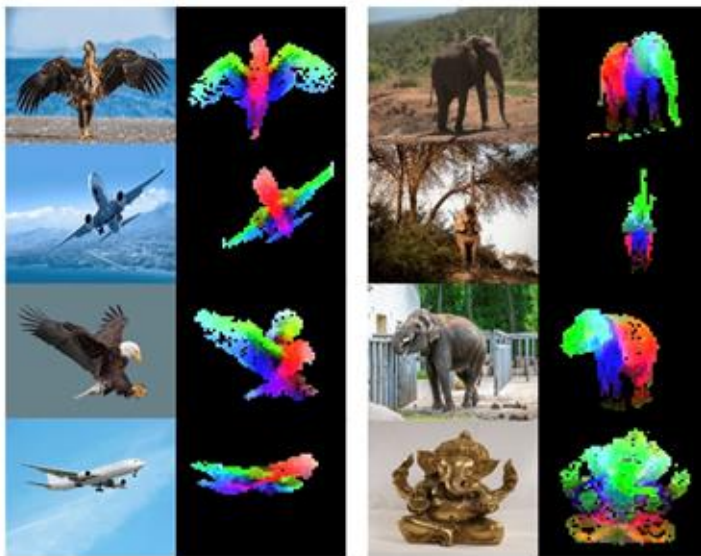


Pretraining



Background
Dataset

Example: Dino V2



<https://dinov2.metademolab.com/>
<https://arxiv.org/abs/2304.07193>

Fine-Tuning of LLMs



Training Data



```
def incr_list(l: list):  
    """Return list with elements incremented by 1.  
    >>> incr_list([1, 2, 3])  
    [2, 3, 4]  
    >>> incr_list([5, 3, 5, 2, 3, 3, 9, 0, 123])  
    [6, 4, 6, 3, 4, 4, 10, 1, 124]  
    """  
    return [i + 1 for i in l]
```

<https://arxiv.org/abs/2107.03374>

(Simplification)

Aligning Different Models

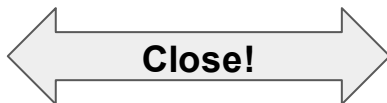


Image
Embedding

“A photo of a television studio.”



Text
Embedding



Aligning Different Models

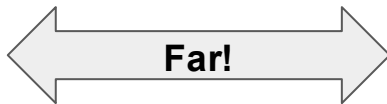


Image
Embedding

“A photo of a lecture room.”



Text
Embedding



Contrastive Learning



Image
Embedding
A

Define Loss on: $A^T C > A^T B$

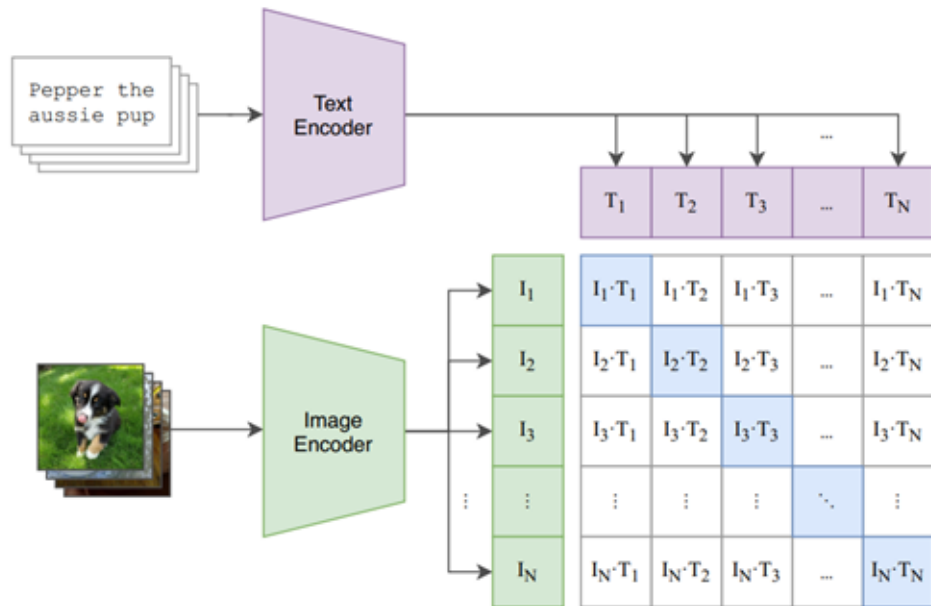
“A photo of a lecture room.”

Text
Embedding
B

Text
Embedding
C

“A photo of a television studio.”

Alignment via Contrastive Learning (CLIP)



```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter
```

```
# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]
```

```
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)
```

```
# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)
```

```
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

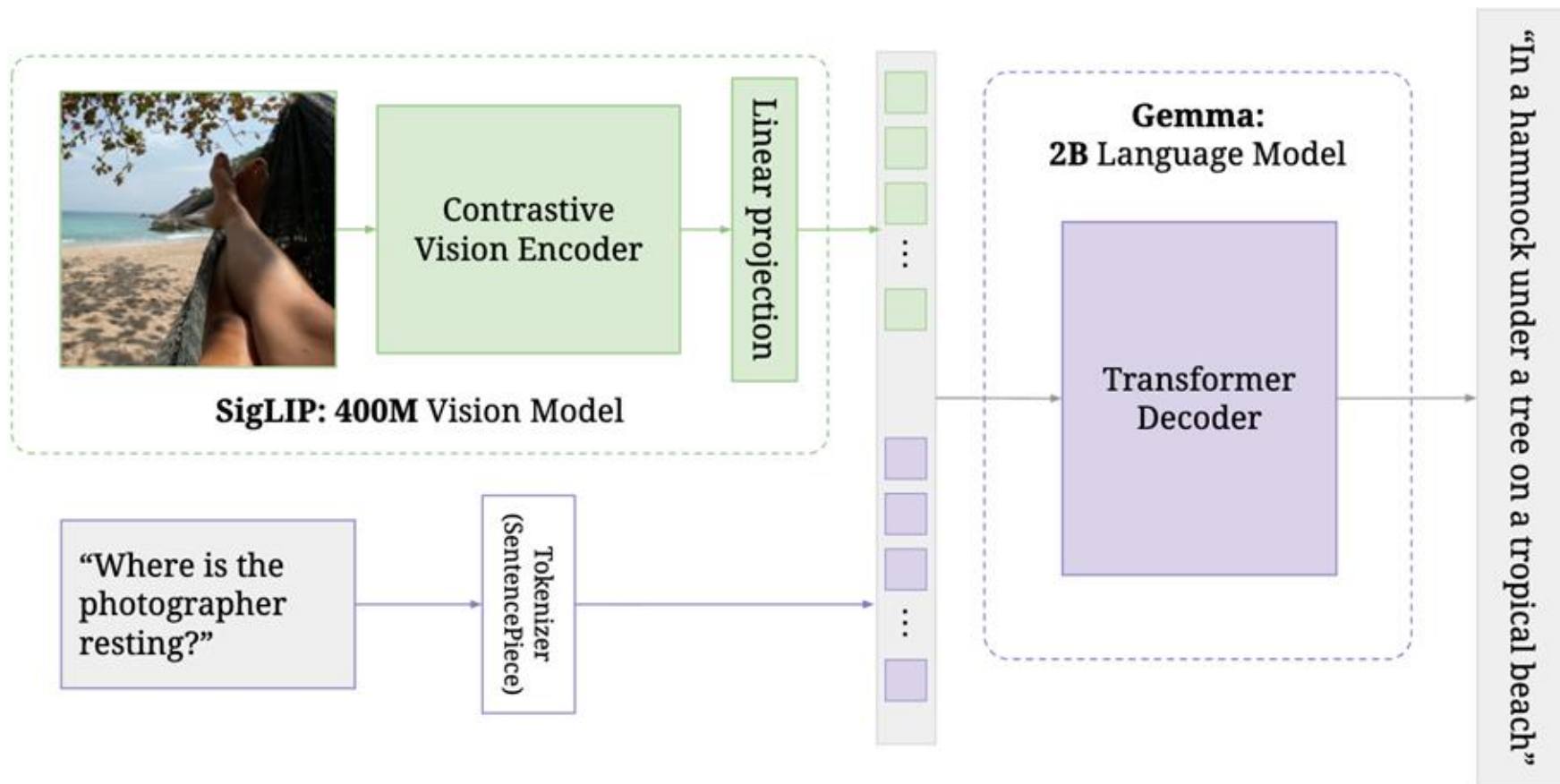
Composite Models



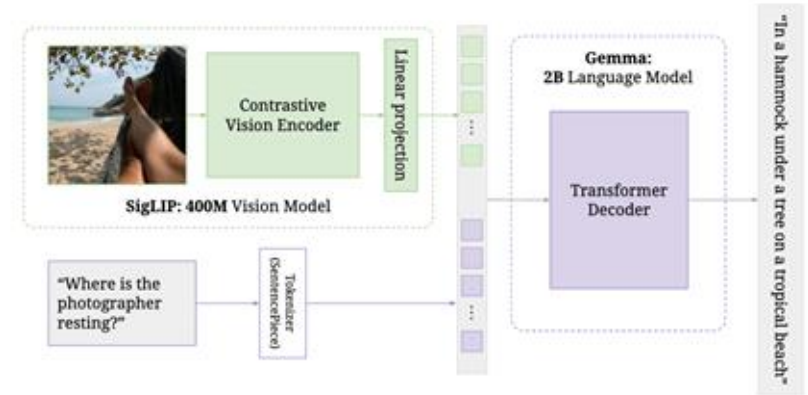
**Text Only
Model**

How to get Gemma to understand images?

PaliGemma



Training Example

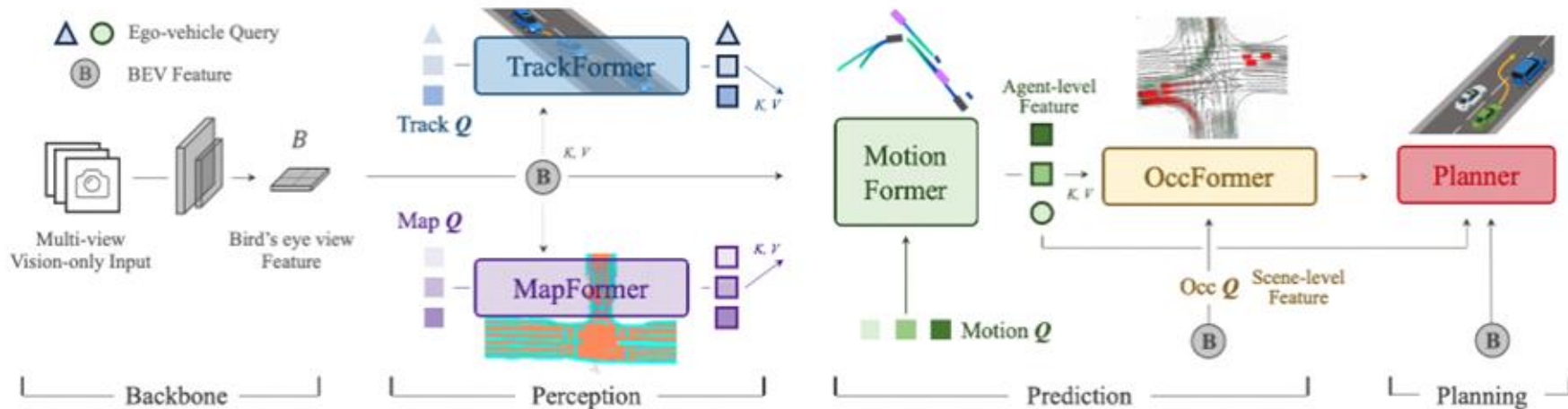


Prefix: "answer en What is a sound this animal makes?\n"

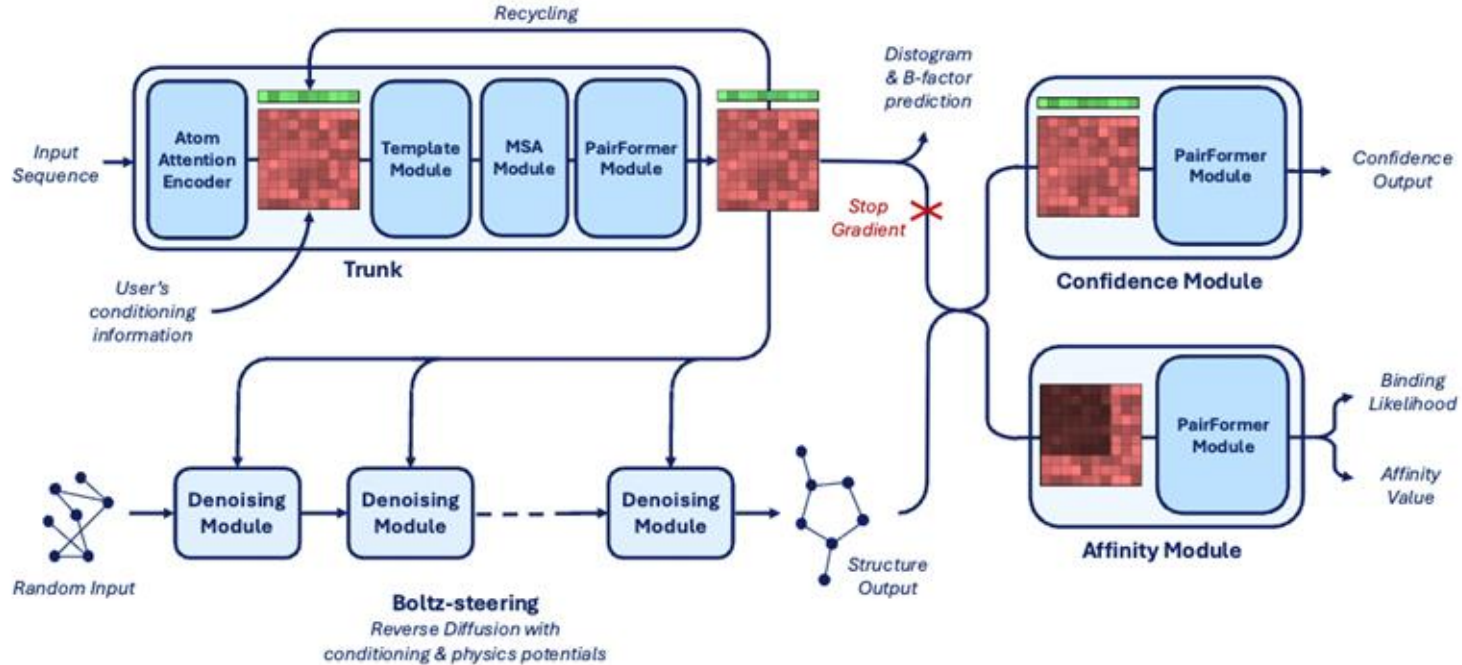
Suffix: "meow"

Train: 18201 examples (train+val splits).

Composite Models are Everywhere!



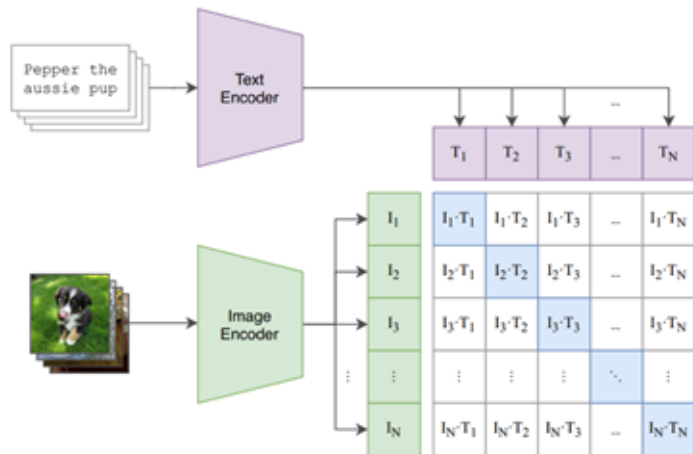
Composite Models are Everywhere!



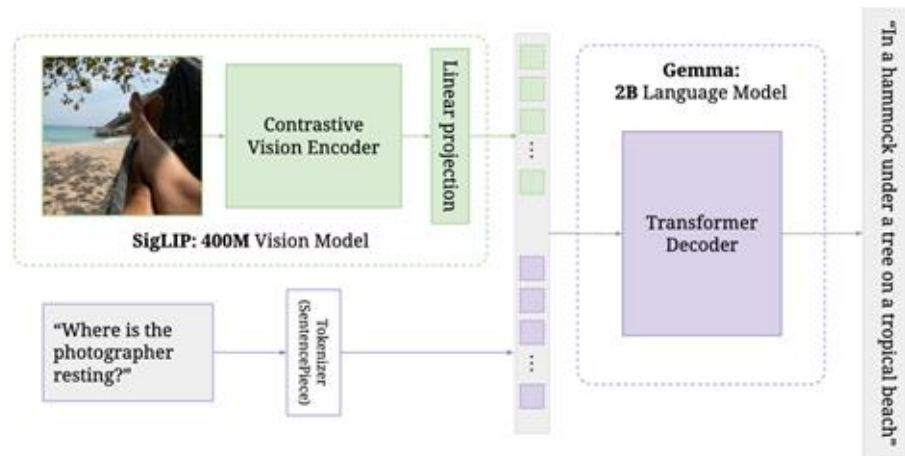
Story so far



Supervised Fine-tuning

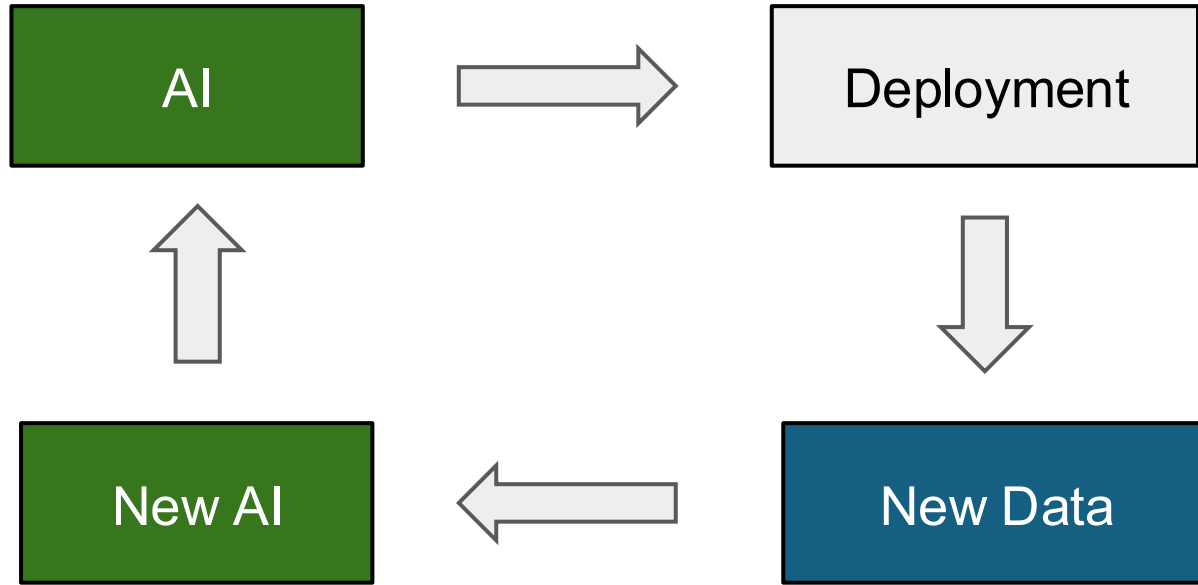


Representation Alignment
(e.g., contrastive learning)

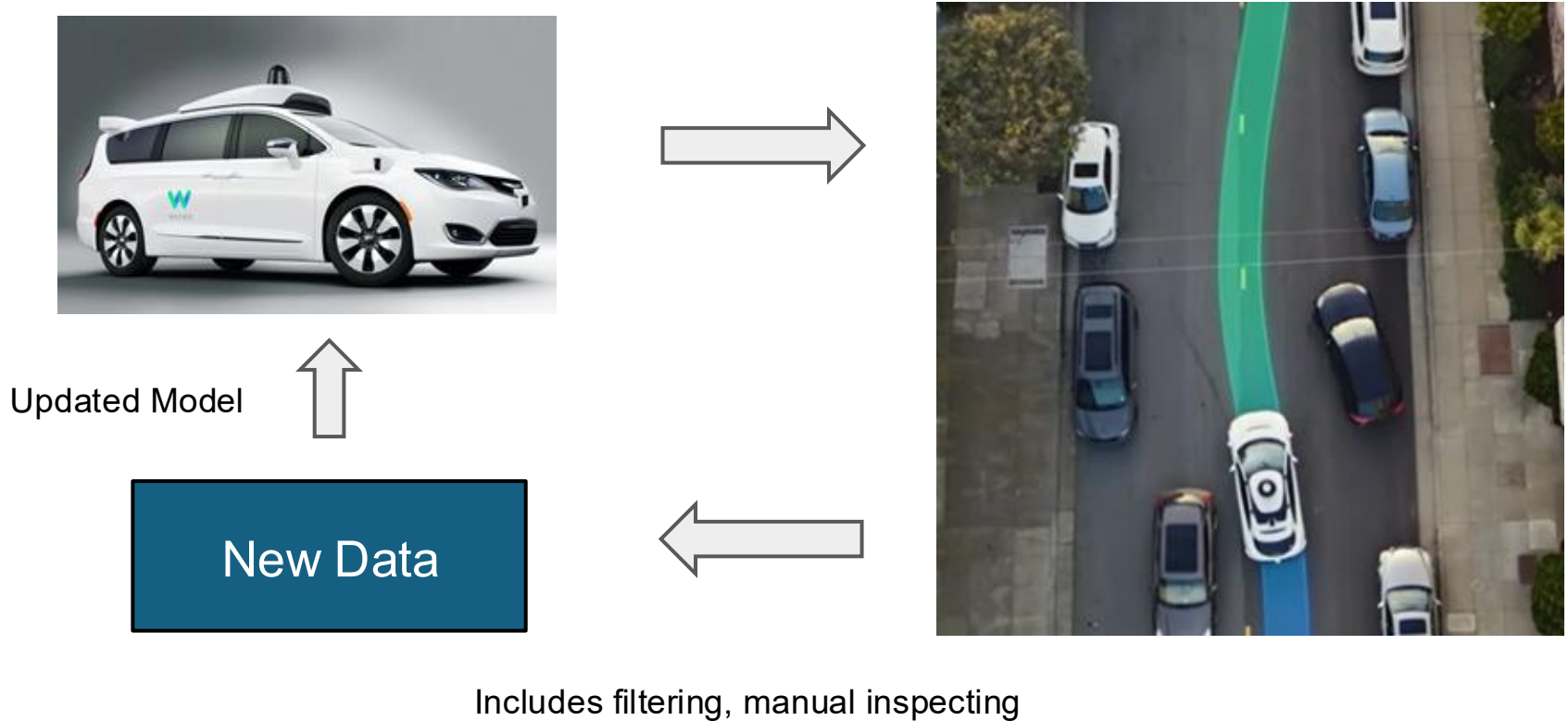


Composite Models

What is a Flywheel?



Flywheels in Autonomous Driving



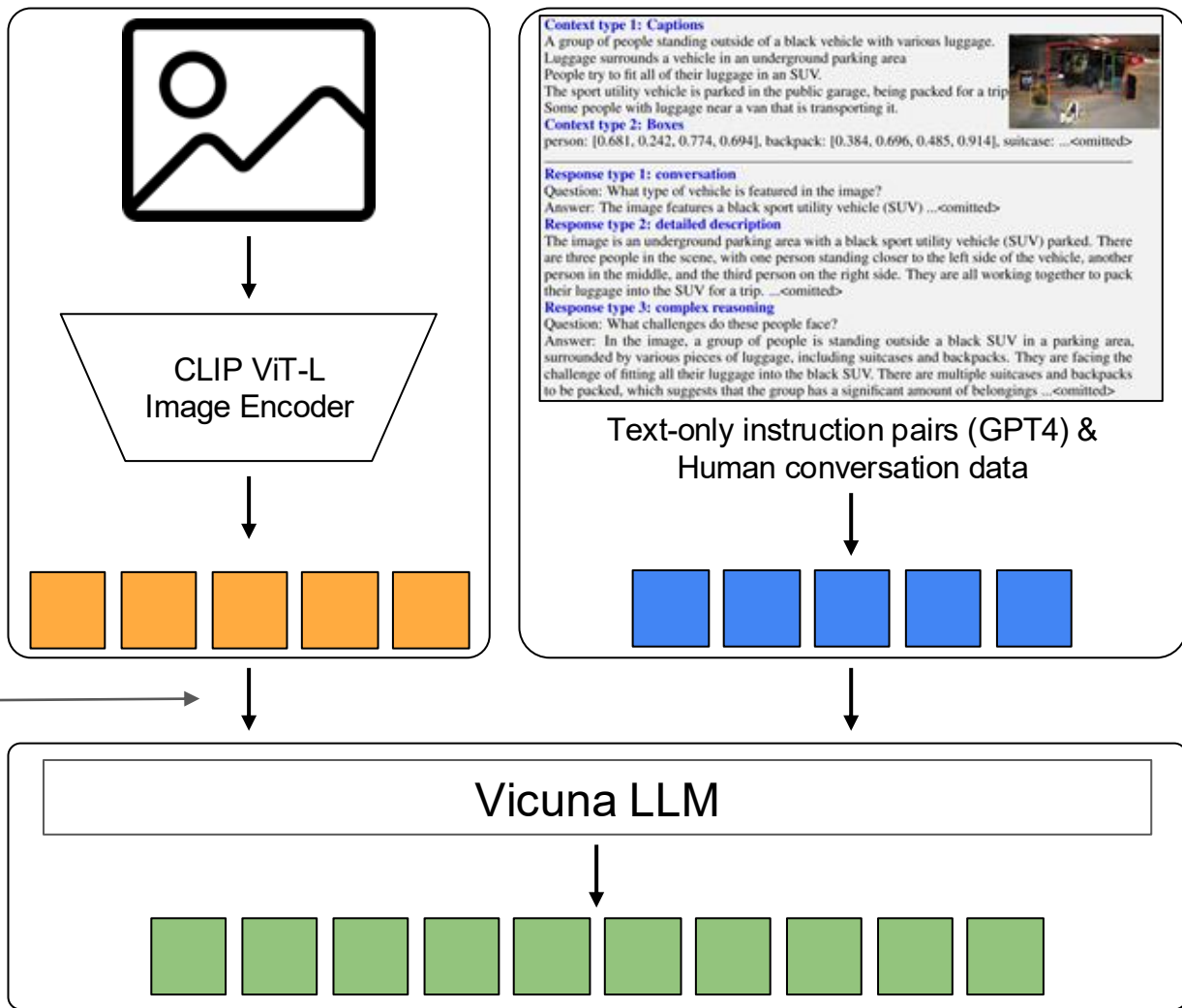
LLaVA

<https://arxiv.org/abs/2304.08485>

Step 1: train CLIP

Step 2: train LLM + adapter
(data from text-only LLM)

“adapter”



LLaVA

<https://arxiv.org/abs/2304.08485>



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User

What is unusual about this image?

LLaVA

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

User

What is unusual about this image?

GPT-4 [36]

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

User

What is unusual about this image?

BLIP-2

a man is sitting on the back of a yellow cab

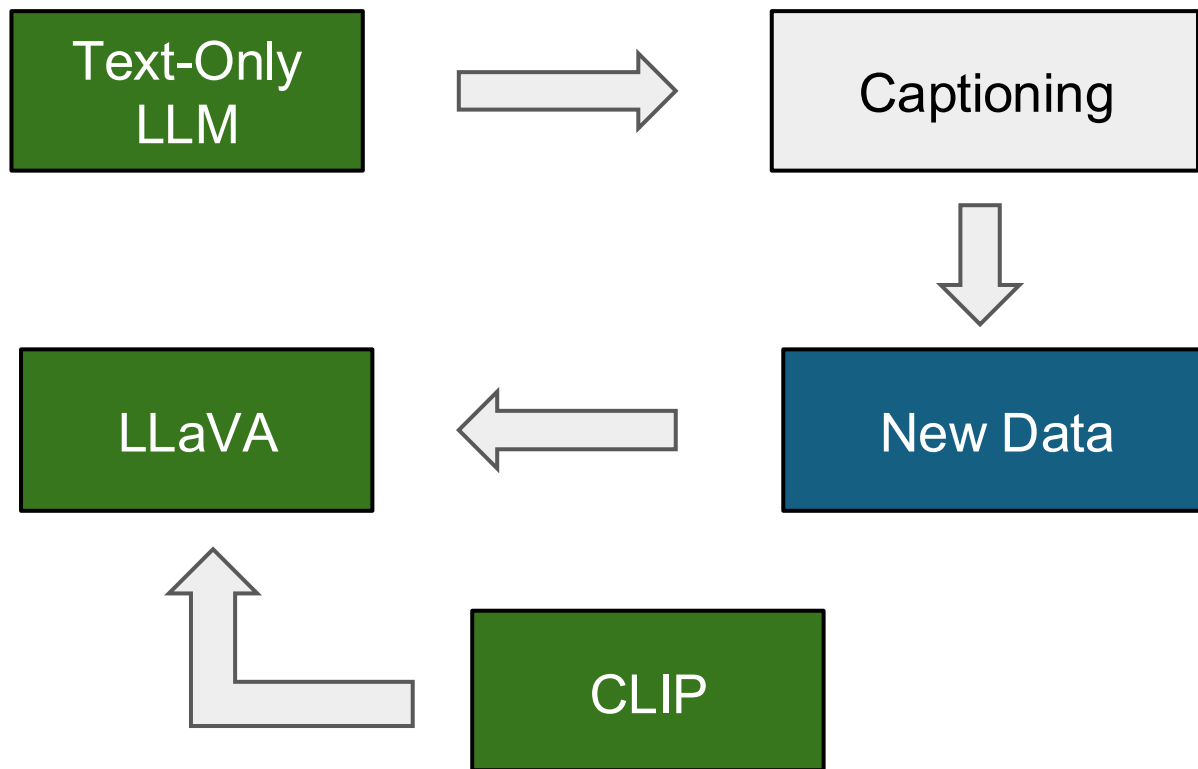
User

What is unusual about this image?

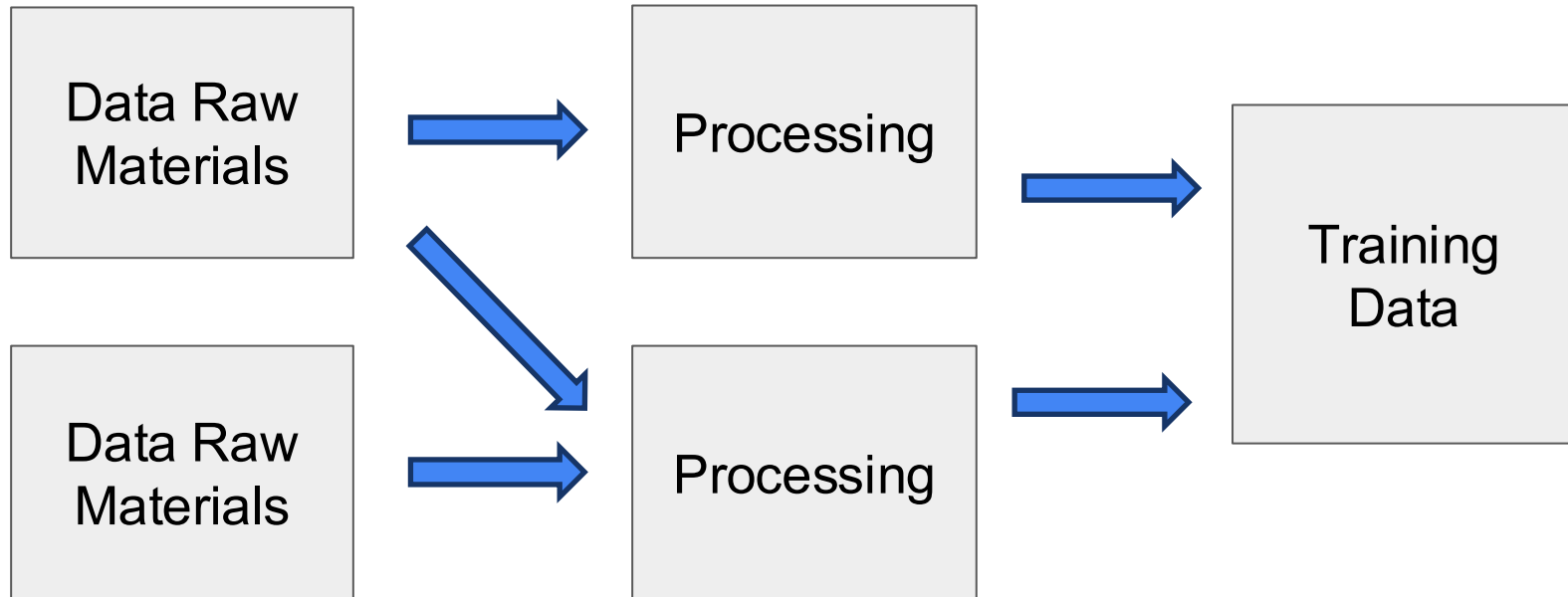
OpenFlamingo

The man is drying his clothes on the hood of his car.

LLaVA = One Turn of Flywheel

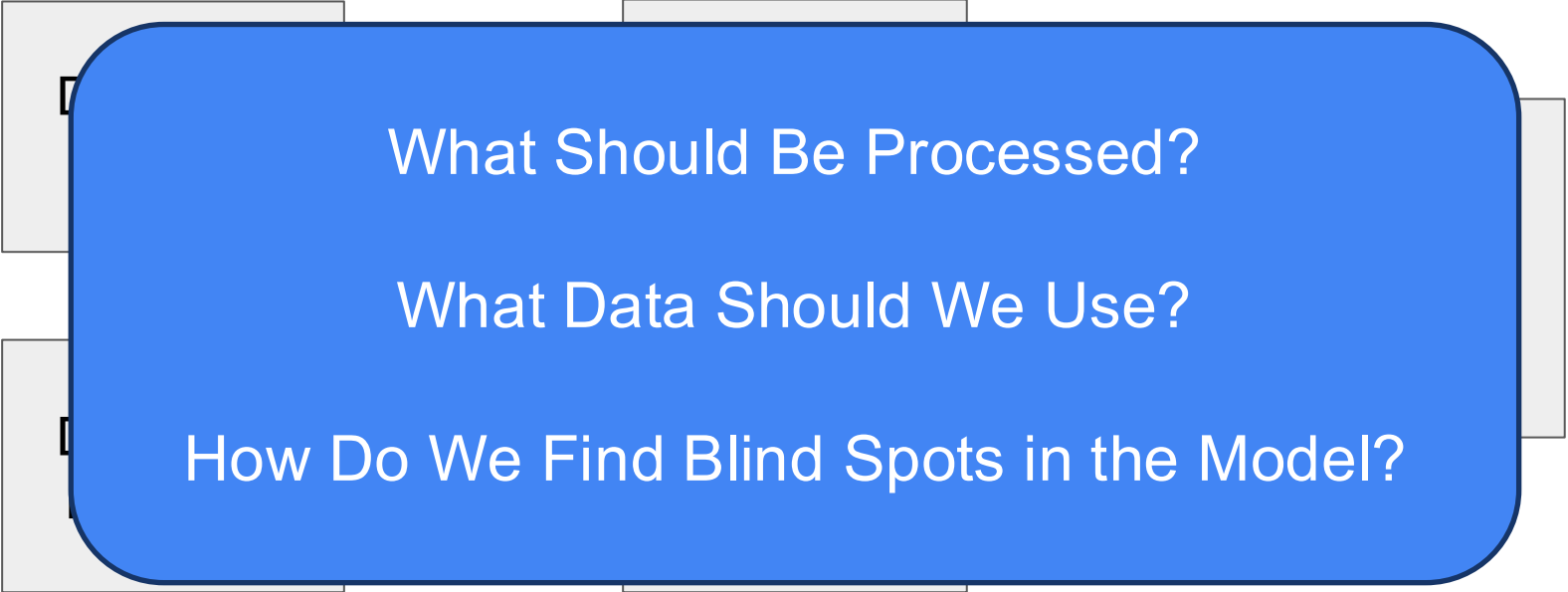


Concept: Data Engines



Scripting, Existing AI, Human Edits

Concept: Data Engines



What Should Be Processed?

What Data Should We Use?

How Do We Find Blind Spots in the Model?

Scripting, Existing AI, Human Edits

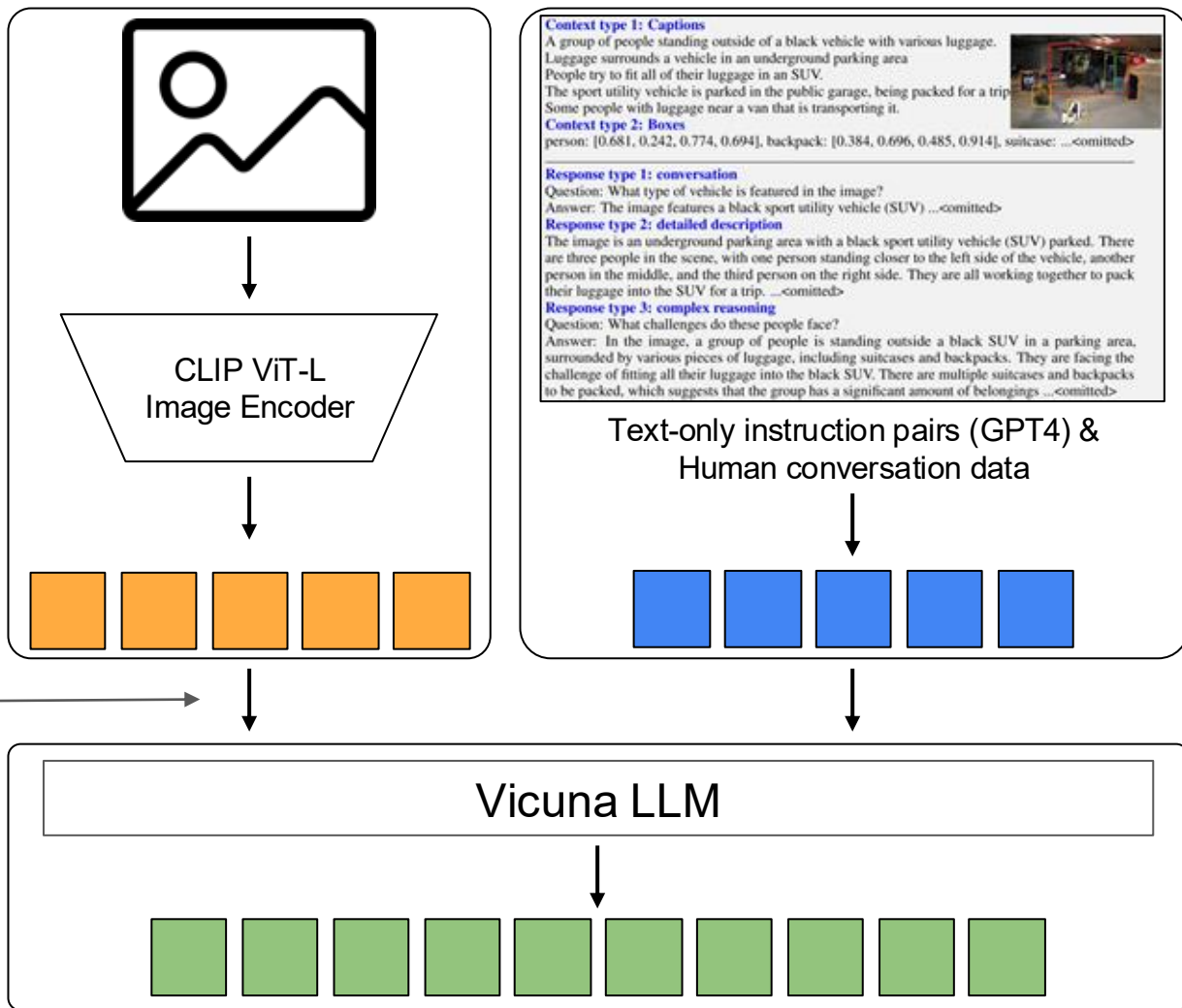
LLaVA Revisited

<https://arxiv.org/abs/2304.08485>

Step 1: train CLIP

Step 2: train LLM + adapter
(data from text-only LLM)

“adapter”



Concept: Jaggedness of AI



x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

99.3 % confidence

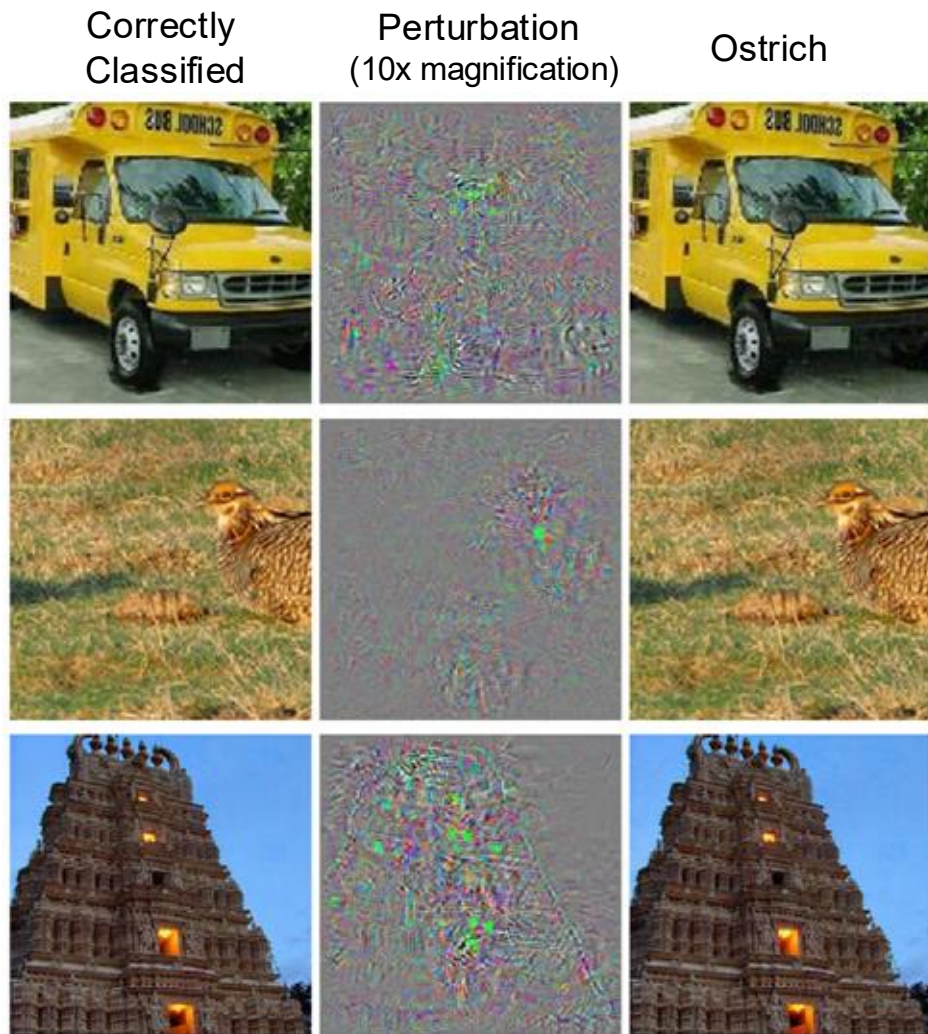
Jaggedness of AI

2013, AlexNet sensitivity analysis

Takeaways:

- sensitivity=anisotropic,
- arbitrarily large misclassification
 - Via worst-case directions in input space

<https://arxiv.org/abs/1312.6199>

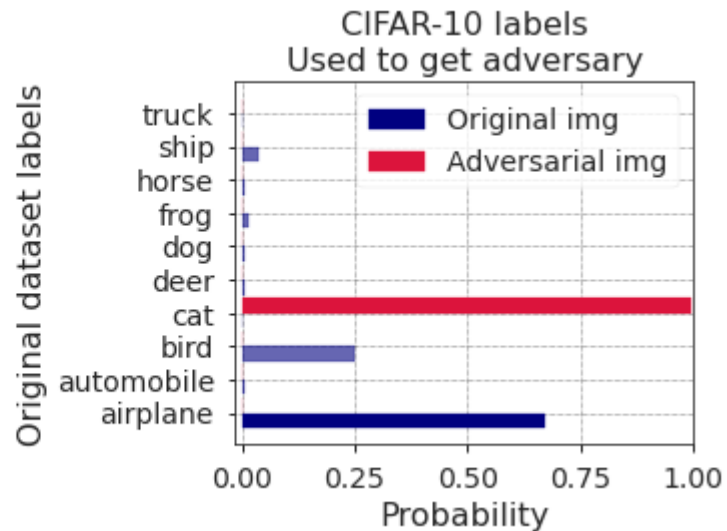


Jaggedness of CLIP

CIFAR-10 original image
"airplane"



Adversarial image
From "airplane" to "cat"



Jaggedness of Image-Text LLMs

 User Query

Which of the following words best describes the theme of this image?



A. Salad
B. Fruit
C. Healthy
D. Chicken

Prompt & Response 

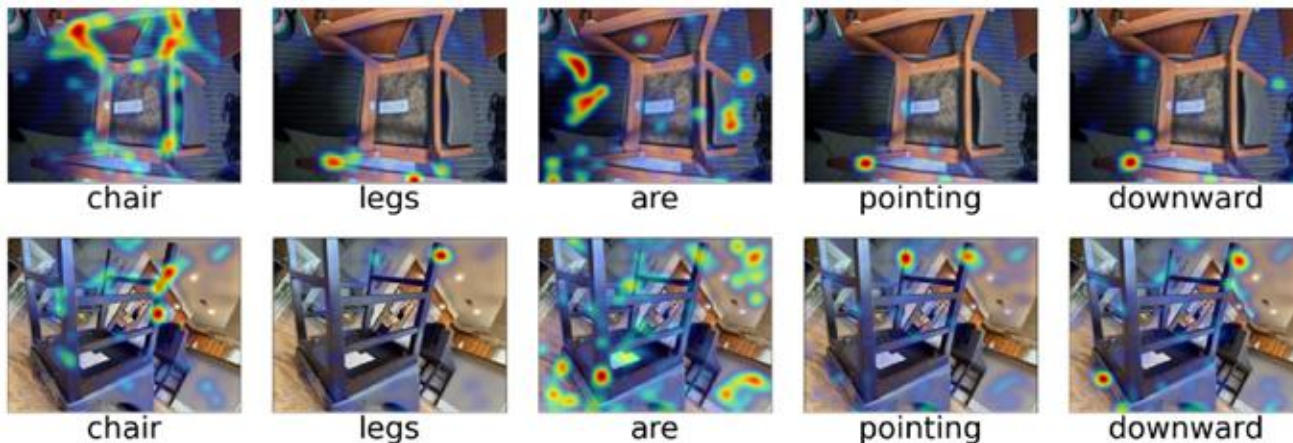
... ****PRIMARY GOAL:**** Focus entirely on the question and the options to determine the correct answer. Your response ****MUST**** be based solely on this information...

C. Healthy 

This question is inherently challenging, and I understand perfection might not be possible. Examine the image carefully and provide the answer letter that you believe best aligns with what you see.

Answer: A 

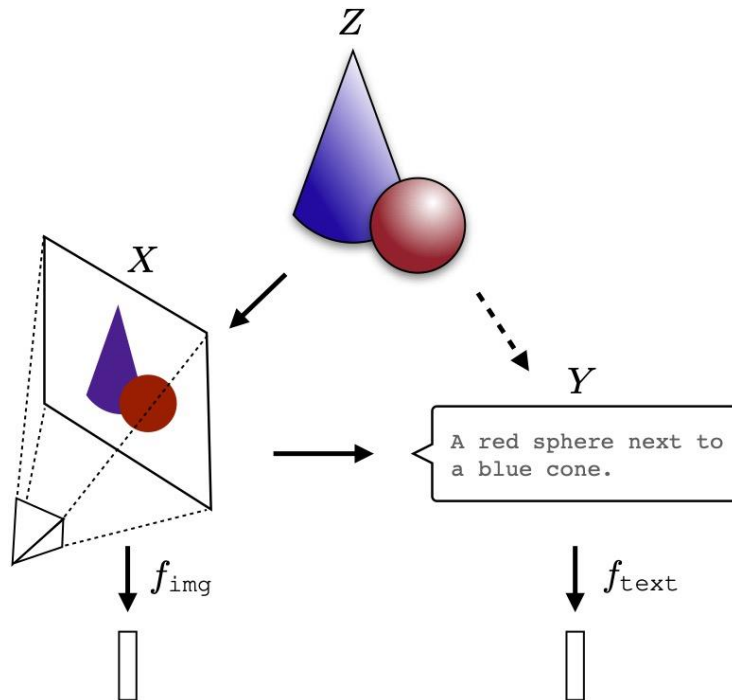
Jaggedness of Image-Text LLMs



“[W]hile image information provided to MLLMs can help performance, they are highly sensitive to perturbations in the image input; often, image information actually hurts performance, indicating weaknesses in the reasoning capabilities of MLLMs”

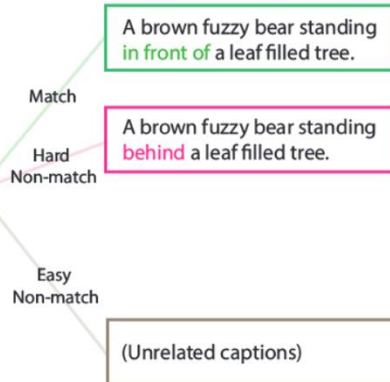
Side Note: Platonic Representation Hypothesis

Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.

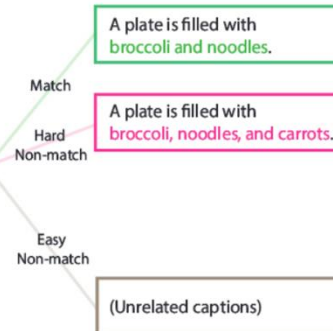


Subtleties in Studying Representational Alignment

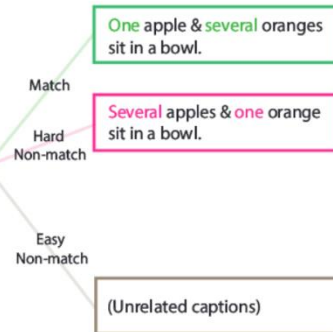
Replace (R) Task



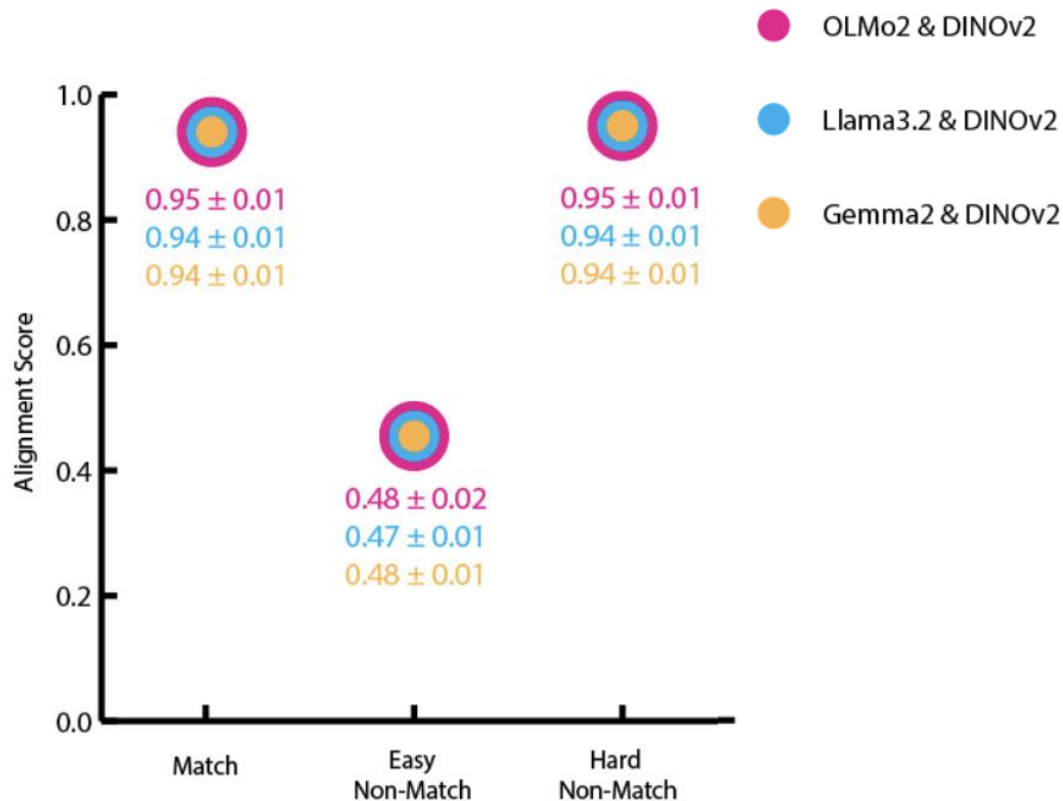
Add (A) Task



Swap (S) Task



Alignment Varies By Nuance!

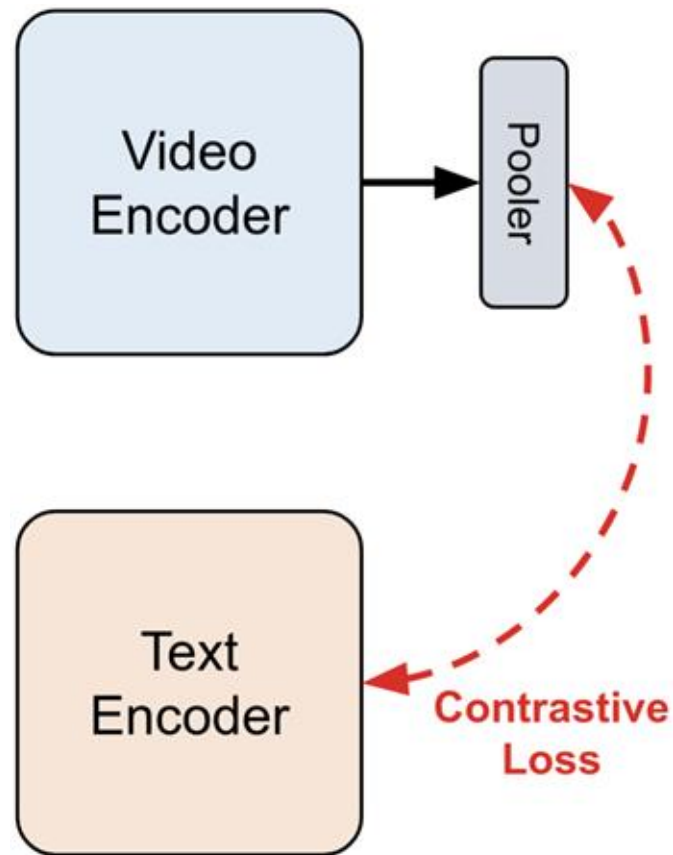


<https://arxiv.org/abs/2507.01201>

VideoPrism

Video-Text Representation Learning

- Trained on internet-scale videos & captions
- Let's analyze its jaggedness!



Example: Single Color Error

Before (P=0.099)	After (P=0.901)
<i>A man is riding a green canoe on a river.</i>	<i>A man is riding a yellow canoe on a river.</i>

Sensitive+Correct!



Example: Long Description Color Error

Before (P=0.388)	After (P=0.612)
<i>A bearded man is paddling a green canoe on a river. The water splashes around the paddle as he steers. Attached to the canoe is a GoPro camera. Behind him is a landscape of flowing water.</i>	<i>A bearded man is paddling a yellow canoe on a river. The water splashes around the paddle as he steers. Attached to the canoe is a GoPro camera. Behind him is a landscape of flowing water.</i>

Moderately Sensitive!



Example: Multi-Color Error

Before (P=0.499)	After (P=0.501)
<i>A man is riding a red canoe on a river with yellow paddles.</i>	<i>A man is riding a yellow canoe on a river with red paddles.</i>

Insensitive!



Example: Action Error

Before (P=0.653)	After (P=0.347)
<i>Paddling</i> a canoe.	<i>Floating still</i> in a canoe.

Moderately Sensitive+Right!



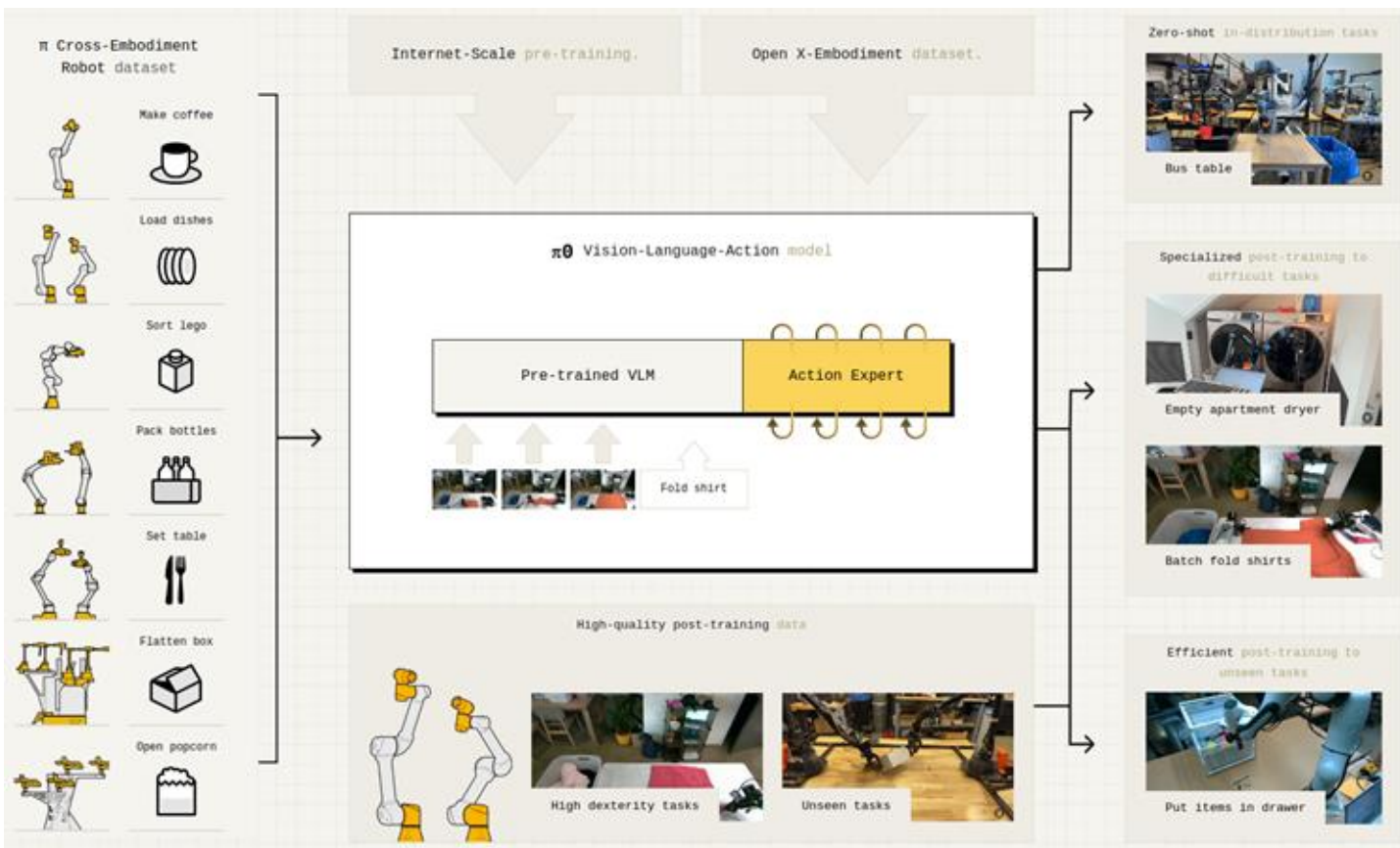
Example: Fine-Grained Specificity

Before (P=0.388)	After (P=0.612)
<i>Gopro</i>	<i>Recording</i>

Moderately Sensitive!



Vision-Language Models in Robotics?



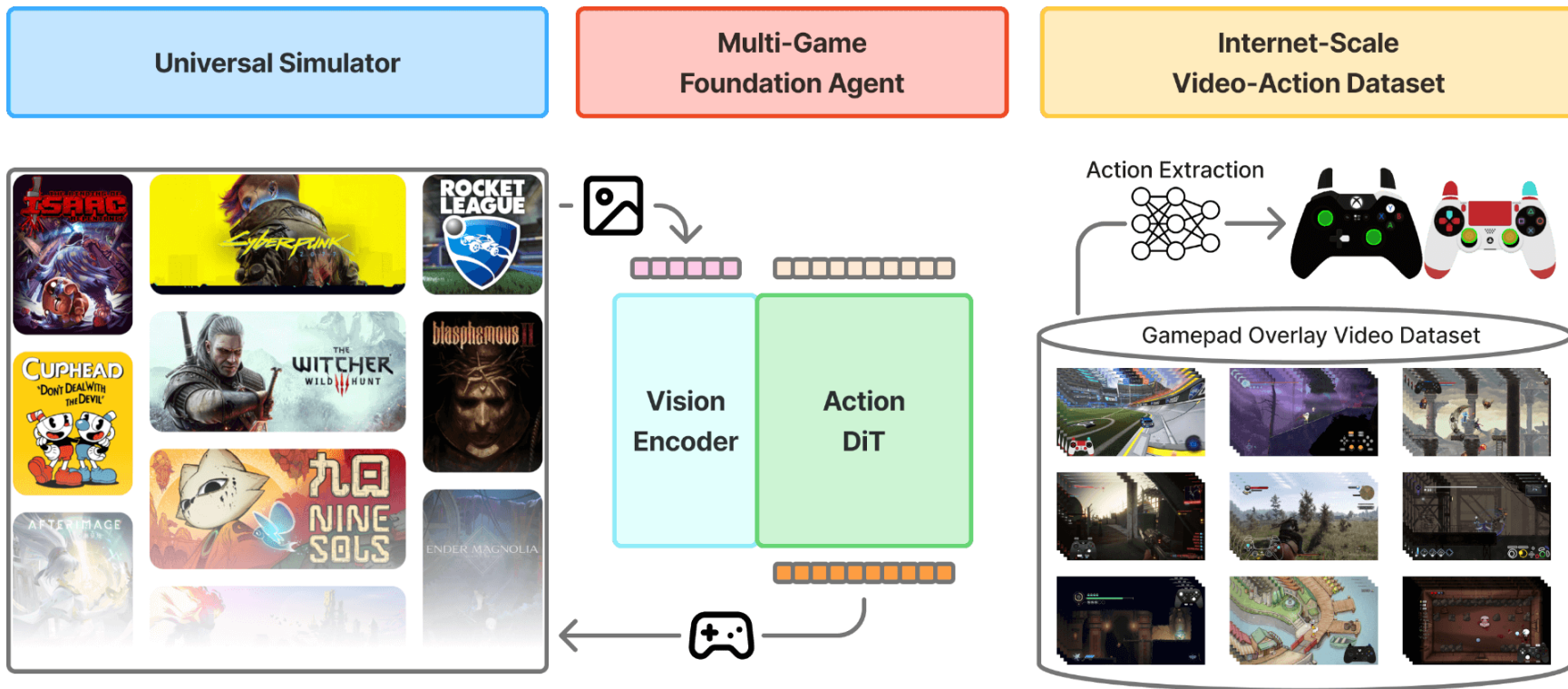
Vision-Language Models in Robotics

Similar Challenges

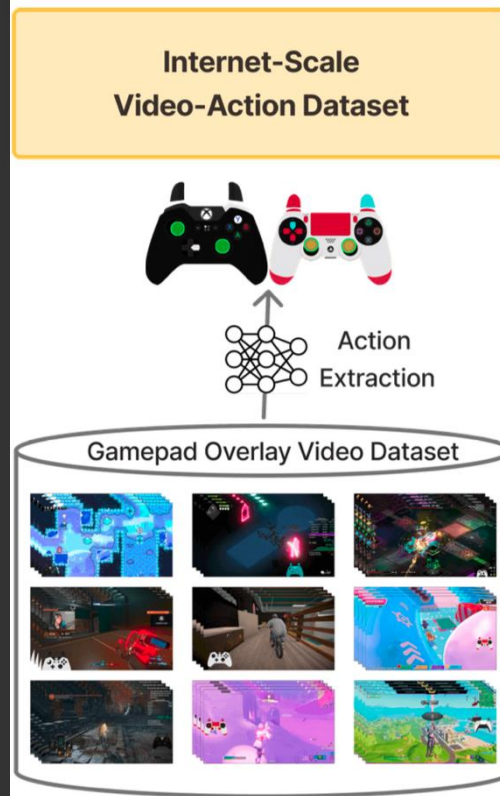
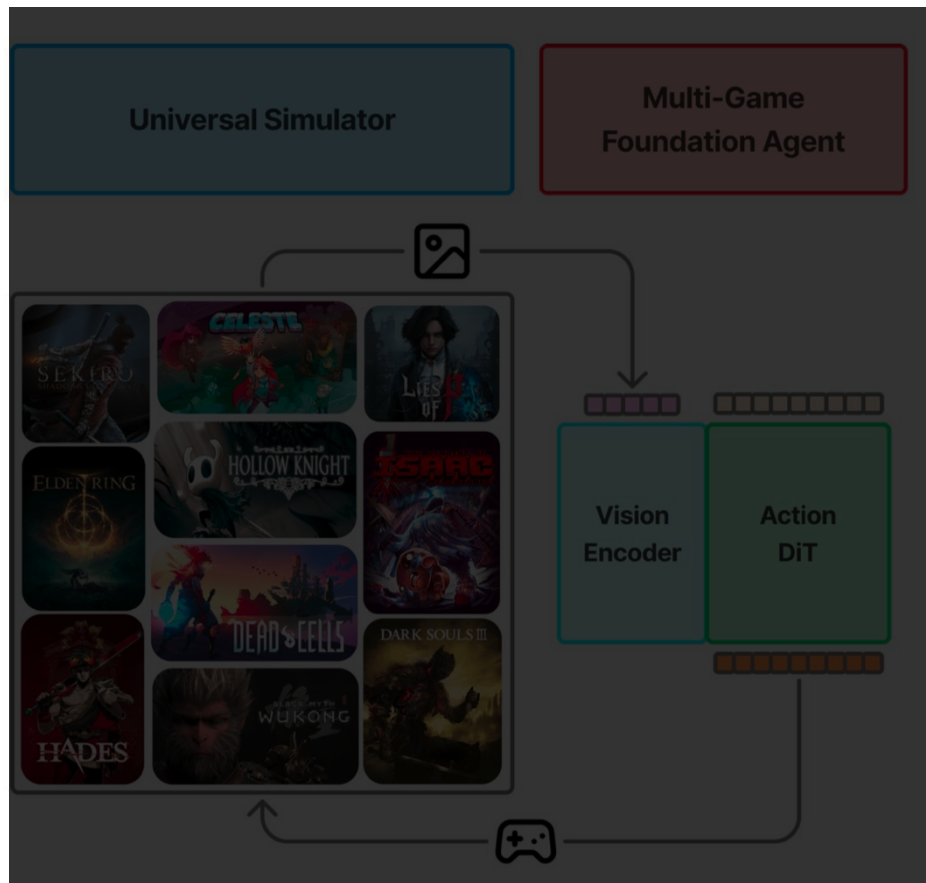
- Demonstrations must “cover” task distribution
- Correcting errors in the human demonstrations
- Overcoming insensitivity of the action expert to human error



NitroGen



NitroGen Data Engine







9:21 AM / CLEAR







CONTRACT: THE CREATURE FROM
OXENFURT FOREST

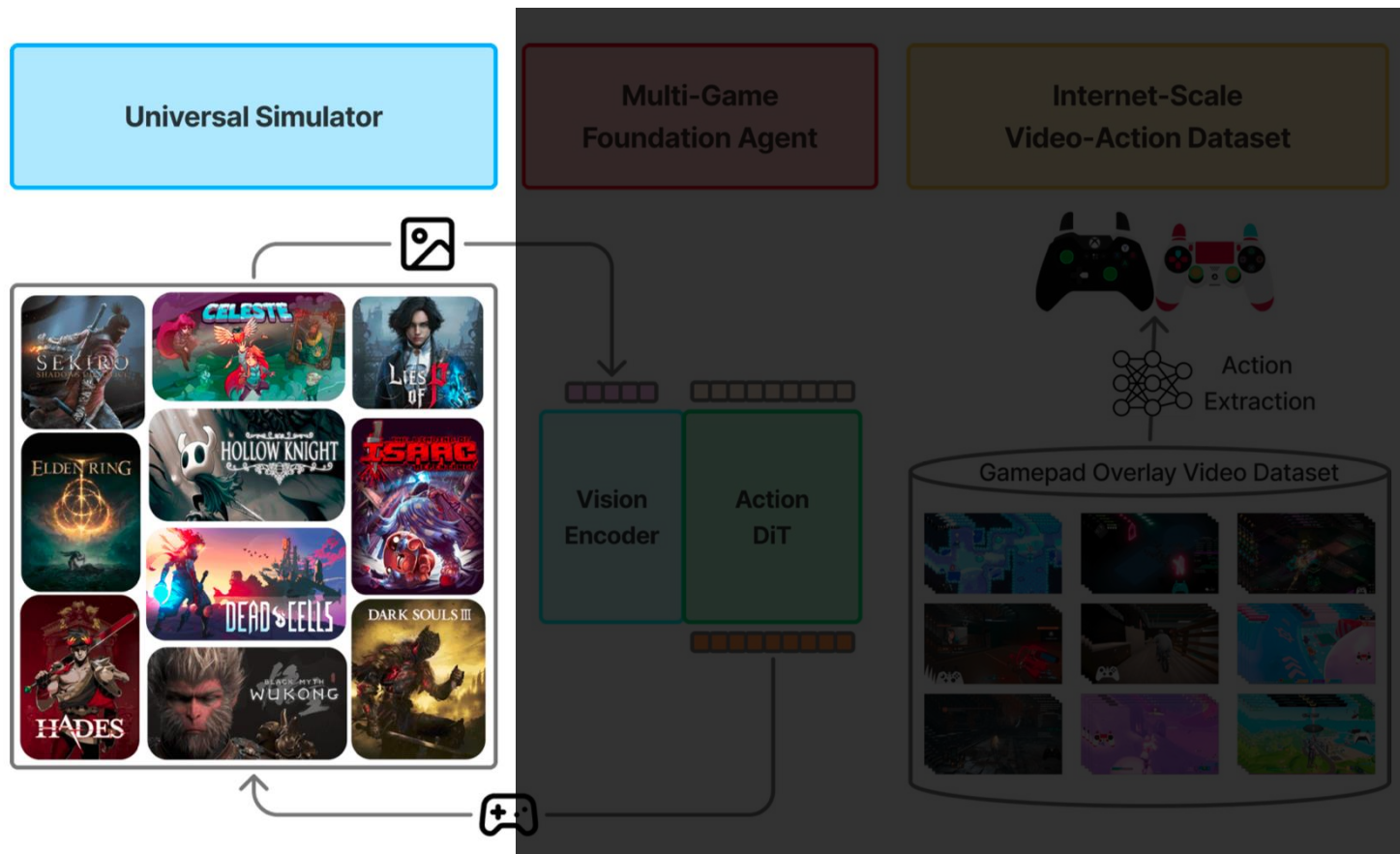
Examine where the monster attacked
using your Witcher Senses.

- 4 Enhanced Thunderbolt
- 5 Superior Swallow
- 2 Northern Wind

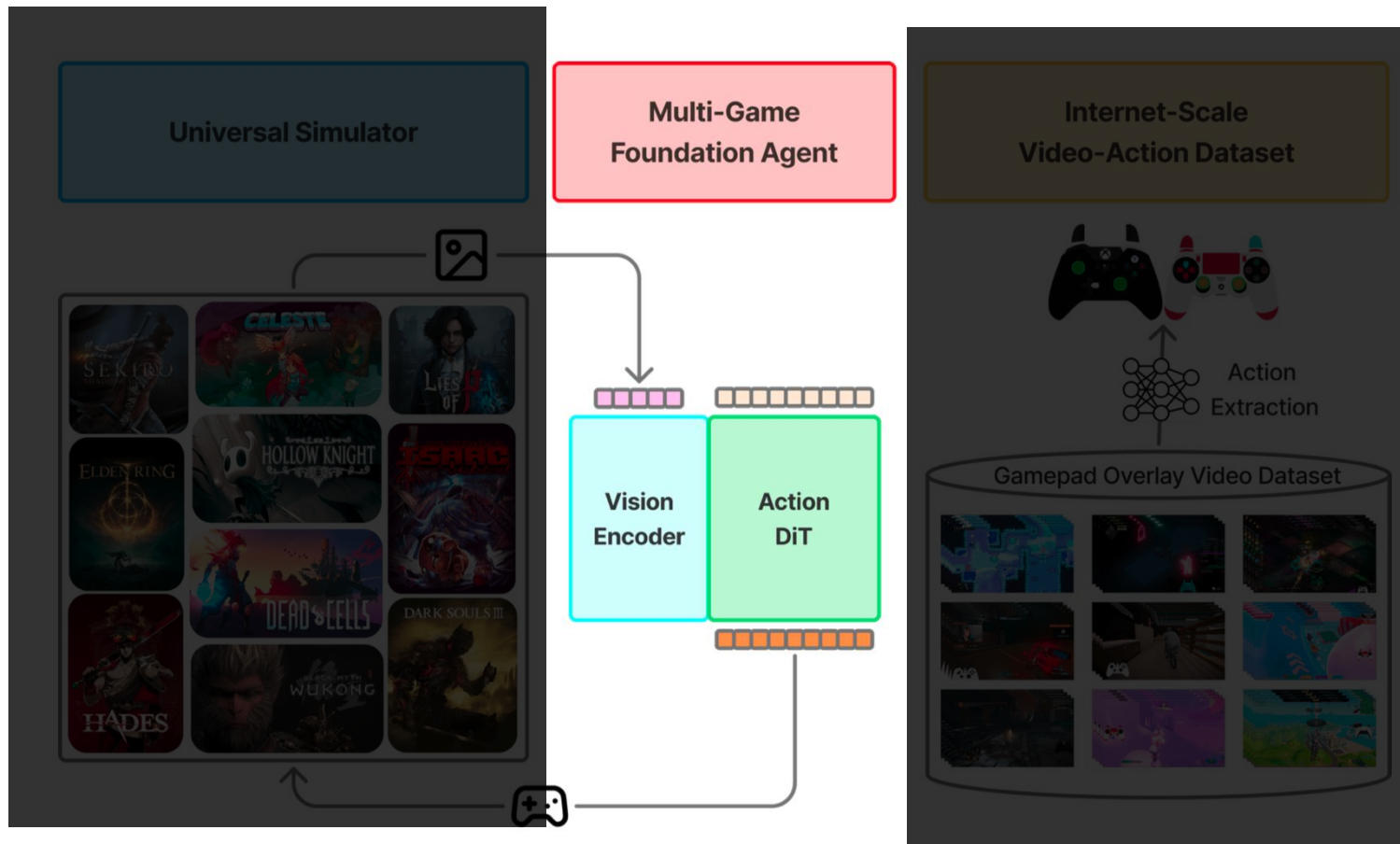


- Silver sword
- Fast Attack 
- Strong Attack 
- Dodge 
- Cast Sign 

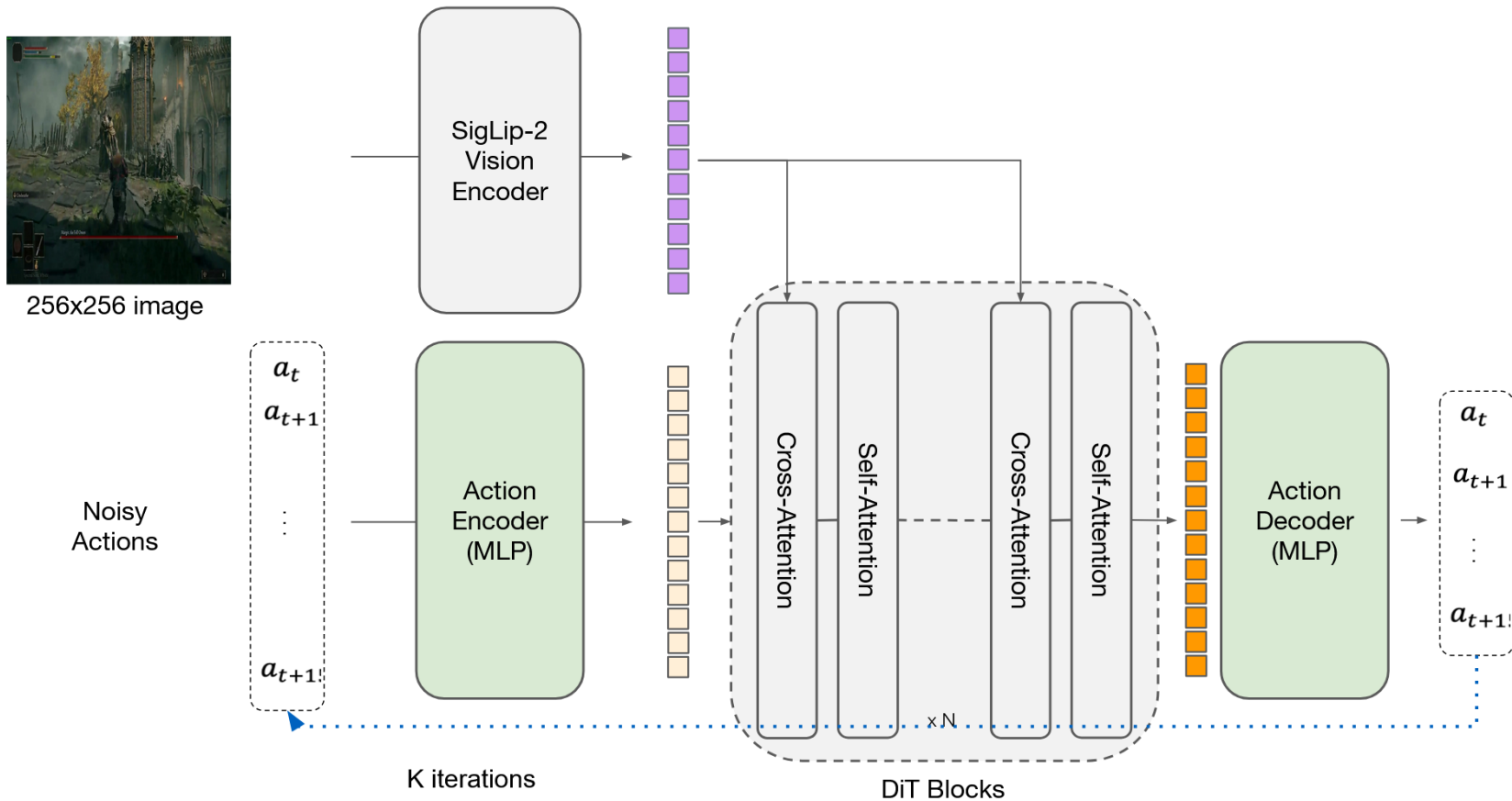
NitroGen Evaluation



NitroGen Model



NitroGen Model

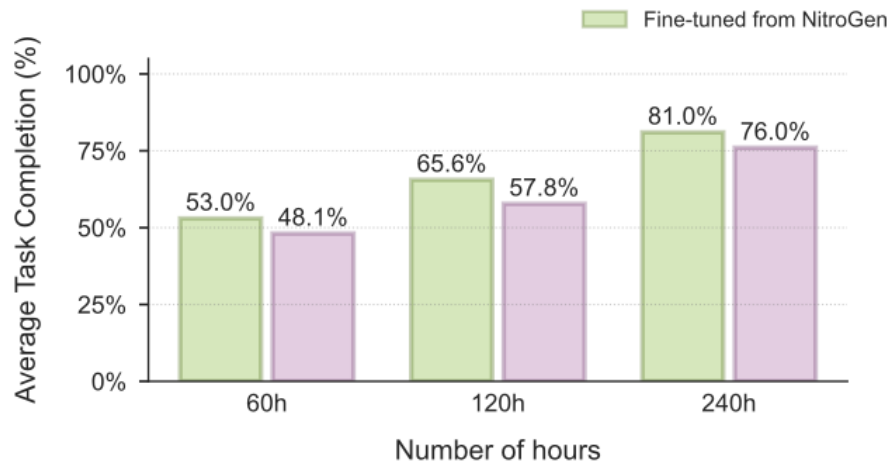


Autonomous agent

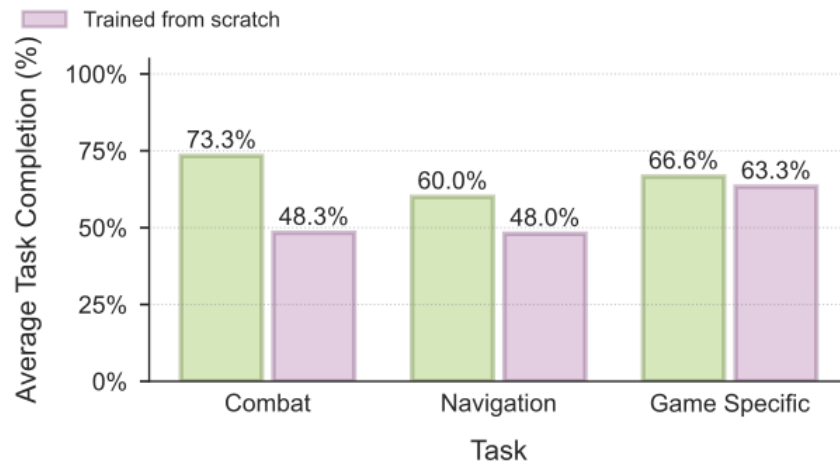


Fast flip to reach the ball first

Quantitative Results



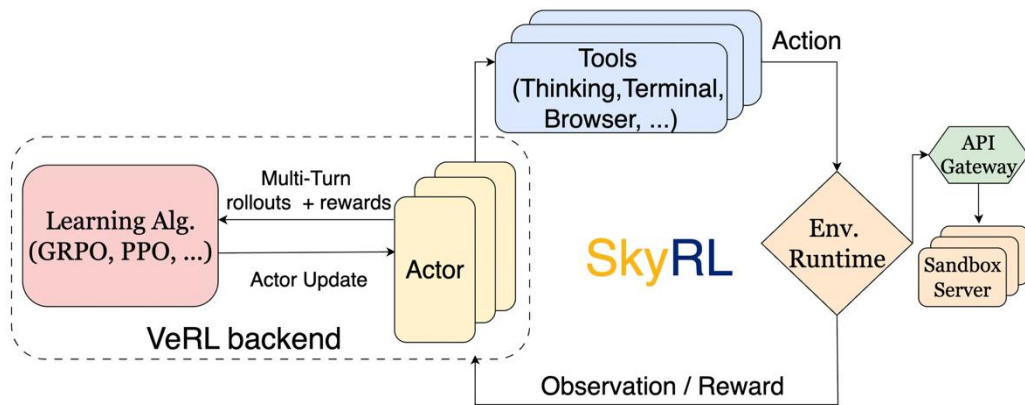
(a) Fine-tuning vs. from scratch:
varying data quantity (isometric roguelike game)



(b) Fine-tuning vs. from scratch:
varying task types (3D action-RPG, 30h data)

Flywheels in AI Agents

- Most of Claude Code is written by CC (same w/ Codex)
- Can train LLMs within Agentic Workflows



<https://arxiv.org/abs/2511.16108>



Summary

- AI models build on each other
- AI models used in data engines
- Design choices in how to smooth out jaggedness
- Design choices in how to align
- **Result: AI Flywheels**